

Intervals and absolute value equations

Exact solution hulls for inverse M-matrix intervals

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: A polynomial hull algorithm would need to exploit the inverse-M promise beyond the known endpoint reductions, warranting challenging. Its direct impact is on a specialist class of verified linear-system computations.

Area: verified linear systems; interval complexity

Independently reviewed resolution - 2026-09-11

Affirmative algorithmic resolution. Theorem 3 and Sections 3-5 give the exact coordinatewise solution hull using $2n$ rational LPs, each with n variables and $2n$ inequalities, in polynomial binary input length. The construction uses precisely the inverse-M promise and arbitrary interval right-hand sides. Promise recognition is not needed; no solver for the general regular AV-03 problem is claimed.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the [complete manuscript](#), [independent agent review](#) and [submission record](#). The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. This is agent verification, not external human peer review or formal proof-assistant certification.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Problem statement

For arbitrary $n \geq 1$, the input consists of rational matrices $L \leq U$ and rational vectors $l \leq u$, with entrywise inequalities. Let

$$\mathcal{A} = \{A \in \mathbb{R}^{n \times n} : L \leq A \leq U\}, \quad \mathcal{b} = \{b \in \mathbb{R}^n : l \leq b \leq u\}.$$

Assume as a promise that every $A \in \mathcal{A}$ is an inverse M-matrix: A^{-1} has nonpositive off-diagonal entries and is a nonsingular M-matrix. Here a nonsingular M-matrix means an invertible real matrix with nonpositive off-diagonal entries and entrywise nonnegative inverse. The entries of A and b vary independently.

Is there a deterministic algorithm that returns the exact rational endpoints of

$$\prod_{i=1}^n \left[\min_{A \in \mathcal{A}, b \in \mathcal{b}} (A^{-1}b)_i, \max_{A \in \mathcal{A}, b \in \mathcal{b}} (A^{-1}b)_i \right]$$

in time polynomial in the binary input length? Behavior outside the promise is unrestricted, and checking the promise is not part of the requested algorithm. This is the precise bit-complexity formulation of the source's efficient interval-system solution-hull question.

The promise makes the solution set nonempty and compact. The aim is its smallest coordinatewise enclosure. Computing the interval hull of A^{-1} separately is already easier in this matrix class; multiplying that enclosure by $\mathcal{b}$ need not capture the dependence among entries of the inverse.

References

Milan Hladík, *An overview of polynomially computable characteristics of special interval matrices*, in *Beyond Traditional Probabilistic Data Processing Techniques*, Springer (2020), 295–310; [arXiv:1711.08732v1](#), §9, p. 11, question immediately before Theorem 25. Theorem 24 concerns the different inverse-matrix hull; Theorem 25 specifies the right-hand-side vertices attaining each endpoint.

Status check

On 2026-09-10, searches for inverse M-matrix interval systems, exact solution hull, polynomial/efficient computation, and 2025/2026 found no resolution. The author's publication list through 2026 and Garloff–Al-Saafin–Adm's [2021 interval-property paper](#), §4, were checked. The latter addresses recognizing the matrix class, not this promised hull computation. The explicit openness statement remains historical.

Independent audit — 2026-09-10

The full Hladík preprint, §9, asks the efficient hull question immediately before Theorem 25. That theorem fixes the optimizing right-hand-side vertices but leaves the matrix optimization unresolved; it is not a polynomial algorithm or a solved subclass of this target. The 2021 interval-property paper addresses recognition instead. Later searches specific to inverse M-matrix solution hulls found no resolution; the explicit open-status evidence remains historical.